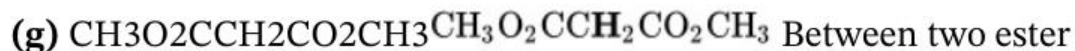
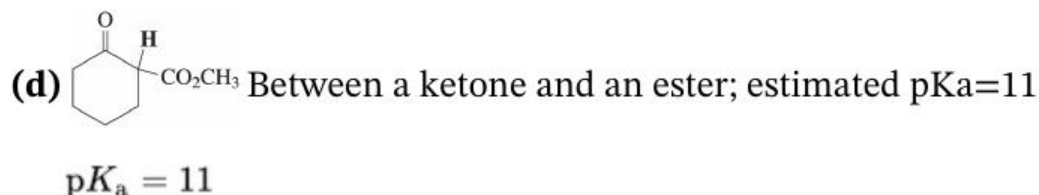
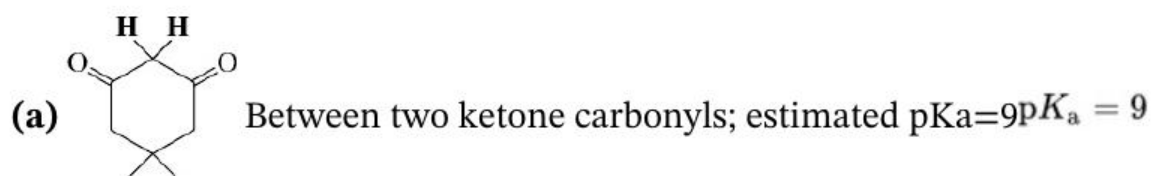


Solutions to Problems

27. Begin by identifying any functional group that contains a hydrogen on a heteroatom (such as oxygen) and assigning to it (or just looking up) a reasonable pK_a value: **(b)** $\text{CH}_3\text{CO}_2\text{H}$, $\text{CH}_3\text{CO}_2\text{H}$, 4.8 (Table 19-3); **(c)** CH_3OH , CH_3OH , 15.5 (Table 8-2). *Sooner or later (preferably sooner) you should memorize the facts that simple carboxylic acids typically have pK_a values around 4 to 5, and alcohols have pK_a values around 16 to 18.*

Next, look at all compounds for which the most acidic hydrogen is attached to an α -carbon of a carbonyl group. Especially, identify the ones flanked by multiple carbonyl groups: they will be more acidic. Estimate (or, again, look up—Table 23-1) pK_a values. Be sure to distinguish between the types of carbonyl functions (aldehyde, ketone, ester, etc.), because they differ in how much they acidify the α -hydrogens: α -hydrogens: aldehydes acidify the most, esters the least, and ketones are intermediate.

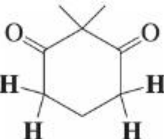
α - α -Hydrogens between two carbonyl groups:



carbonyls; estimated $pK_a = 13$

α -Hydrogens adjacent to only one carbonyl group:

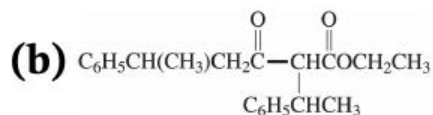
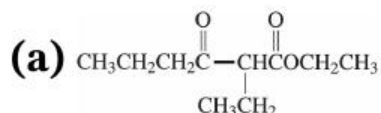
(e) CH_3CHO Aldehyde; estimated $pK_a = 17$

(f)  Ketone; estimated $pK_a = 20$

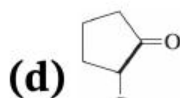
Compound (h) has no α -hydrogens. The hydrogens on its methyl groups are not appreciably acidified. It comes in last. So the final order of acidity is (b) > (a) > (d) > (g) > (c) > (e) > (f) > (h).

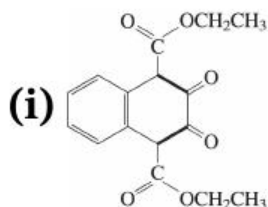
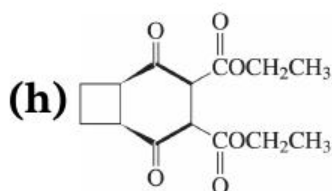
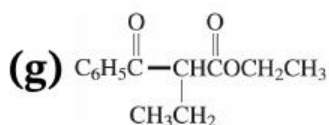
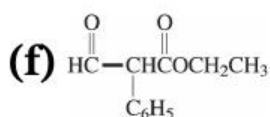
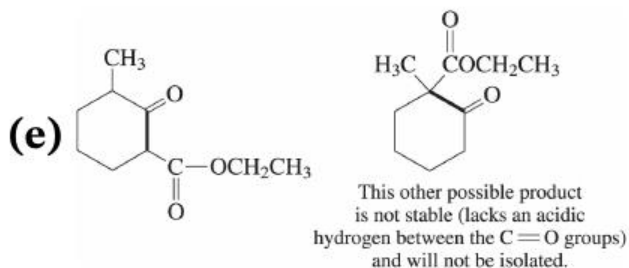
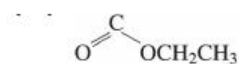
(b) > (a) > (d) > (g) > (c) > (e) > (f) > (h).

28. Claisen condensations. (a), (b), (c) involve two identical molecules; (d), (e) are intramolecular examples; (f), (g), (h), (i) are mixed condensations. Make your new carbon-carbon bond (boldface) between the carbonyl carbon of one ester and the α -carbon of another.

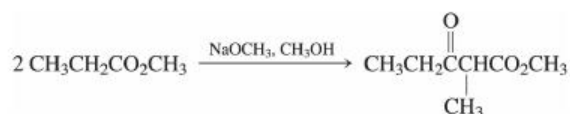


(c) Unfavorable equilibrium, because the α -carbon has only one H on it. Claisen product is not stable, because it lacks an acidic hydrogen that can be removed by base; no reaction is observed.

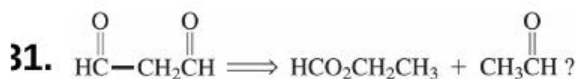
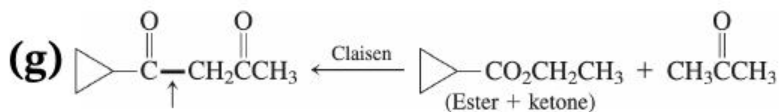
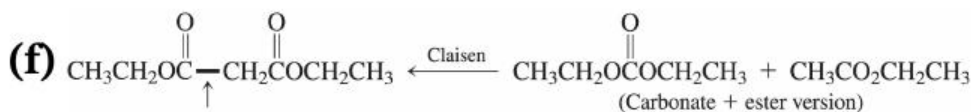
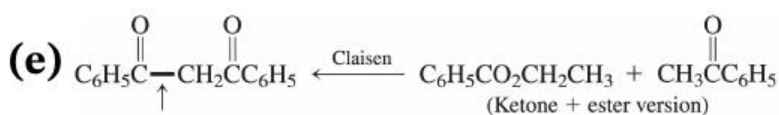
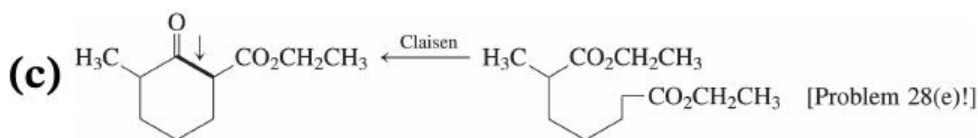
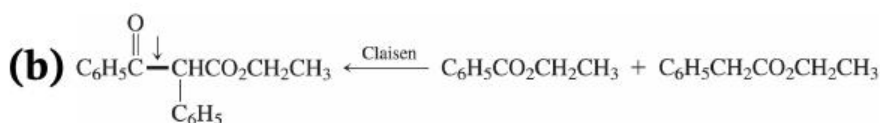
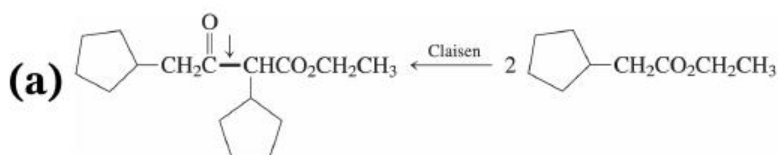




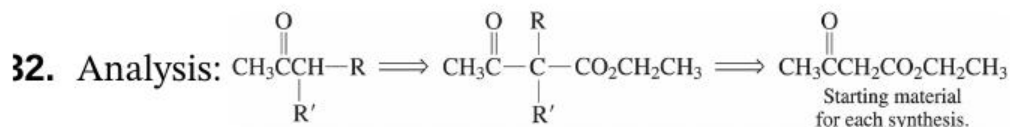
29. The second ester, $(\text{CH}_3)_2\text{CH}-\overset{\text{O}}{\parallel}\text{C}-\text{OCH}_3$, should be present in excess because (1) it does not form a stable product from Claisen condensation with itself and (2) it will be able to preferentially react with enolate ions from the first ester. Side reaction (condensation of first ester with itself):



30. Analyze as you did for [Problem 28](#). “Claisen” means 1.
 $\text{NaOCH}_2\text{CH}_3, \text{CH}_3\text{CH}_2\text{OH}, \text{NaOCH}_2\text{CH}_3, \text{CH}_3\text{CH}_2\text{OH}, 2. \text{H}^+, \text{H}_2\text{O}.$
 $\text{H}^+, \text{H}_2\text{O}.$



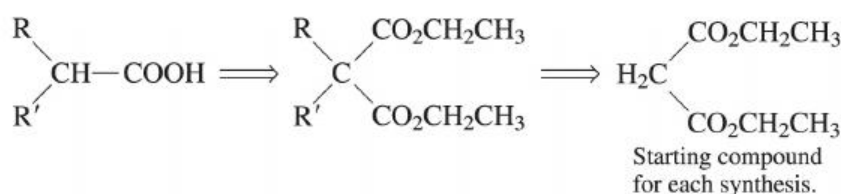
Not likely to work because aldol condensation of 2 CH_3CHO $\xrightarrow{2}$ CH_3CHO would be a major competing process.



Assuming we are starting with ethyl esters, the solvent for each reaction in this and the next problem should be ethanol.

- (a) $\text{R}=\text{—CH}_2\text{CH}(\text{CH}_3)_2, \text{R}'=\text{H}$. $\text{R}=\text{—CH}_2\text{CH}(\text{CH}_3)_2, \text{R}'=\text{H}$. 1. $\text{NaOCH}_2\text{CH}_3$, $\text{NaOCH}_2\text{CH}_3$, 2. $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$, $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$, 3. $\text{NaOH}, \text{H}_2\text{O}$, $\text{NaOH}, \text{H}_2\text{O}$, 4. $\text{H}^+, \text{H}_2\text{O}, \Delta$; $\text{H}^+, \text{H}_2\text{O}, \Delta$;
- (b) $\text{R}=\text{R}'=\text{—CH}_2\text{CH}_2\text{CH}_2\text{—}$. $\text{R}=\text{R}'=\text{—CH}_2\text{CH}_2\text{CH}_2\text{—}$. 1. 2 $\text{NaOCH}_2\text{CH}_3$, 2 $\text{NaOCH}_2\text{CH}_3$, 2. $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{Br}$, $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{Br}$, 3. $\text{NaOH}, \text{H}_2\text{O}$, $\text{NaOH}, \text{H}_2\text{O}$, 4. $\text{H}^+, \text{H}_2\text{O}, \Delta$; $\text{H}^+, \text{H}_2\text{O}, \Delta$;
- (c) $\text{R}=\text{—CH}_2\text{C}_6\text{H}_5, \text{R}'=\text{—CH}_2\text{CH}=\text{CH}_2$. $\text{R}=\text{—CH}_2\text{C}_6\text{H}_5, \text{R}'=\text{—CH}_2\text{CH}=\text{CH}_2$. 1. $\text{NaOCH}_2\text{CH}_3$, $\text{NaOCH}_2\text{CH}_3$, 2. $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$, $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$, 3. $\text{NaOCH}_2\text{CH}_3$, $\text{NaOCH}_2\text{CH}_3$, 4. $\text{H}_2\text{C}=\text{CHCH}_2\text{Br}$, $\text{H}_2\text{C}=\text{CHCH}_2\text{Br}$, 5. $\text{NaOH}, \text{H}_2\text{O}$, $\text{NaOH}, \text{H}_2\text{O}$, 6. $\text{H}^+, \text{H}_2\text{O}, \Delta$; $\text{H}^+, \text{H}_2\text{O}, \Delta$;
- (d) $\text{R}=\text{—CH}_2\text{CH}_3, \text{R}'=\text{—CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$.
 $\text{R}=\text{—CH}_2\text{CH}_3, \text{R}'=\text{—CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$. 1. $\text{NaOCH}_2\text{CH}_3$, $\text{NaOCH}_2\text{CH}_3$, $\text{BrCH}_2\text{CO}_2\text{CH}_2\text{CH}_3$, $\text{BrCH}_2\text{CO}_2\text{CH}_2\text{CH}_3$, 3. $\text{NaOCH}_2\text{CH}_3$, $\text{NaOCH}_2\text{CH}_3$, $\text{CH}_3\text{CH}_2\text{Br}$, $\text{CH}_3\text{CH}_2\text{Br}$, 5. $\text{NaOH}, \text{H}_2\text{O}$, $\text{NaOH}, \text{H}_2\text{O}$, 6. $\text{H}^+, \text{H}_2\text{O}, \Delta$, $\text{H}^+, \text{H}_2\text{O}, \Delta$ (decarboxylates only the COOHCOOH on the α -carbon α -carbon to the ketone), 7. $\text{CH}_3\text{CH}_2\text{OH}, \text{H}^+$, $\text{CH}_3\text{CH}_2\text{OH}, \text{H}^+$ (convert other COOHCOOH group back to ethyl ester)

33. General pattern

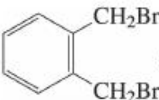


(a) 1. $\text{NaOCH}_2\text{CH}_3$, $\text{NaOCH}_2\text{CH}_3$, 2. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{I}$,
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{I}$, 3. $\text{NaOCH}_2\text{CH}_3$, $\text{NaOCH}_2\text{CH}_3$, 4. - CH_2Br

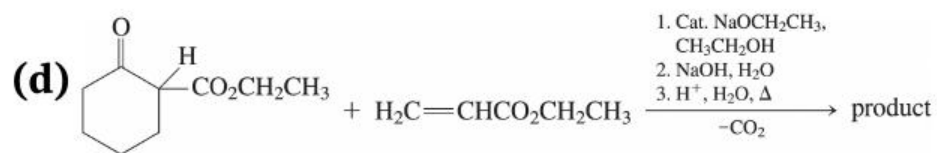
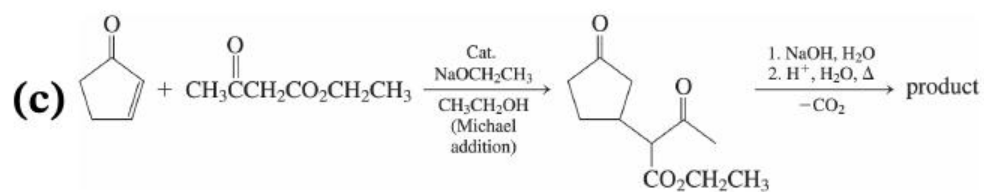
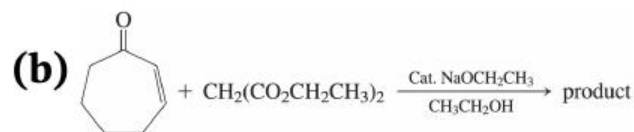
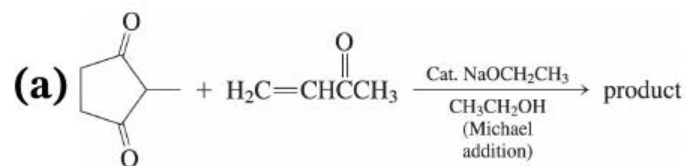
(completes necessary alkylations), 5. NaOH , H_2O , NaOH , H_2O
(hydrolyzes esters), 6. H^+ , H_2O , Δ , H^+ , H_2O , Δ (decarboxylation);

(b) 1. $\text{NaOCH}_2\text{CH}_3$, $\text{NaOCH}_2\text{CH}_3$, 2. $(\text{CH}_3)_2\text{CHCH}_2\text{I}$, $(\text{CH}_3)_2\text{CHCH}_2\text{I}$,
3. $\text{NaOCH}_2\text{CH}_3$, $\text{NaOCH}_2\text{CH}_3$, 4. CH_3I , CH_3I (completes
alkylations), 5. NaOH , H_2O , NaOH , H_2O , 6. H^+ , H_2O , Δ ;
 H^+ , H_2O , Δ ;

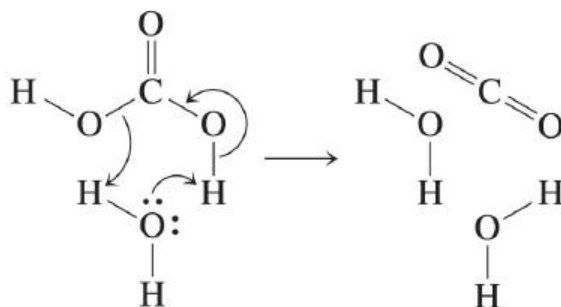
(c) 1. $\text{NaOCH}_2\text{CH}_3$, $\text{NaOCH}_2\text{CH}_3$, 2. $\text{BrCH}_2\text{CO}_2\text{CH}_2\text{CH}_3$
 $\text{BrCH}_2\text{CO}_2\text{CH}_2\text{CH}_3$ [alkylation, makes
 $\text{CH}_3\text{CH}_2\text{O}_2\text{CCH}_2\text{CH}(\text{CO}_2\text{CH}_2\text{CH}_3)_2$,
 $\text{CH}_3\text{CH}_2\text{O}_2\text{CCH}_2\text{CH}(\text{CO}_2\text{CH}_2\text{CH}_3)_2$], 3. NaOH , H_2O , NaOH , H_2O ,
4. H^+ , H_2O , Δ ; H^+ , H_2O , Δ ;

(d) 1. 2 $\text{NaOCH}_2\text{CH}_3$, 2 $\text{NaOCH}_2\text{CH}_3$, 2. , 3. NaOH , H_2O ,
 NaOH , H_2O , 4. H^+ , H_2O , Δ , H^+ , H_2O , Δ

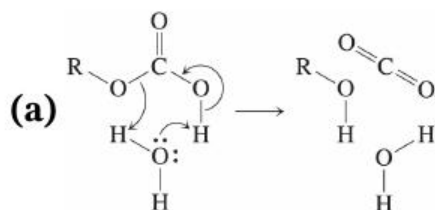
34.

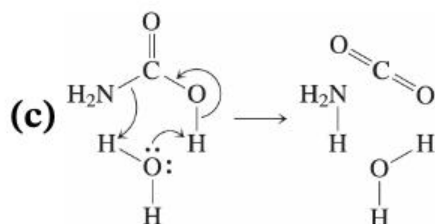


35. Here is one way that gets the job done:

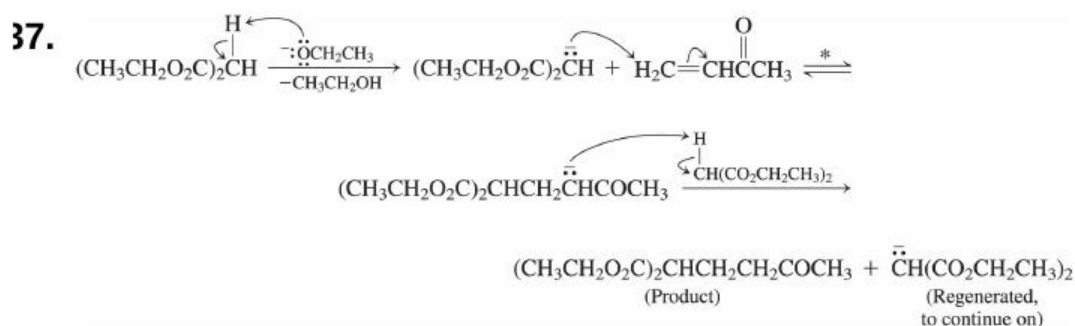


36.



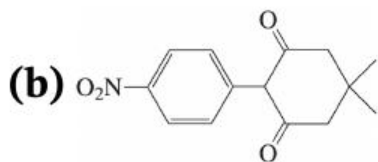
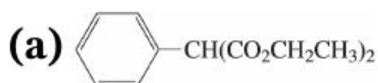


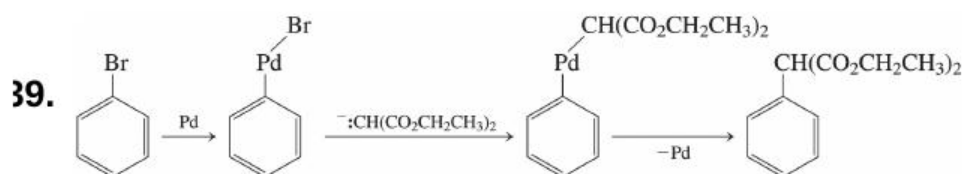
(b), (d) Does not work: An O—H—H bond is necessary (see the mechanisms above).



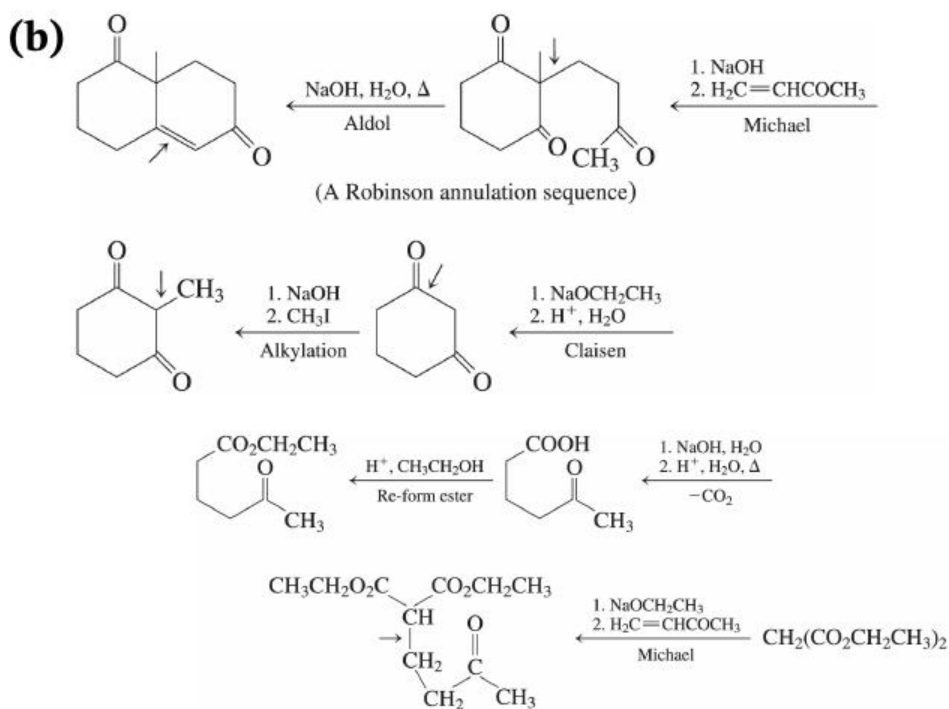
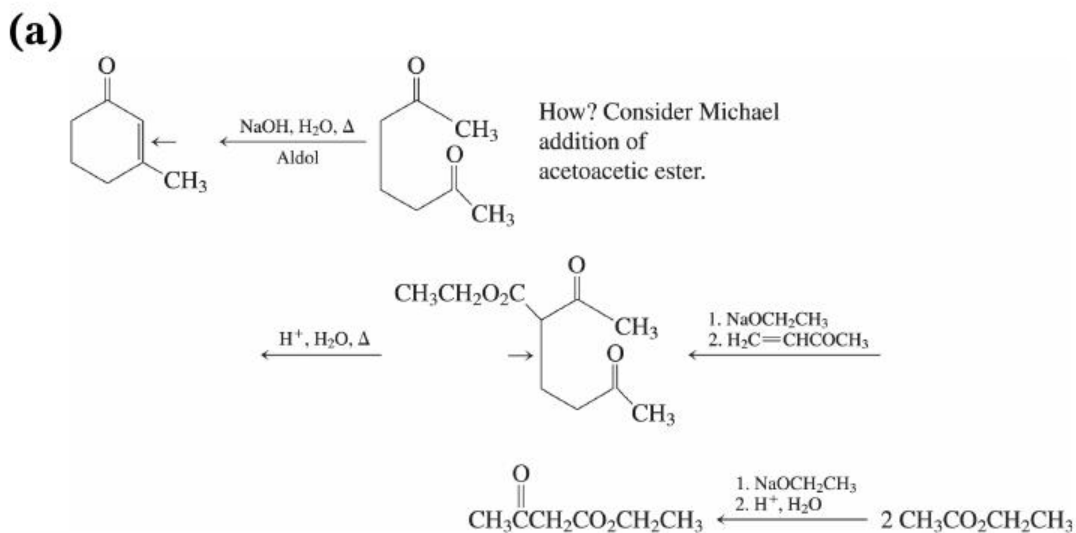
The step marked with an asterisk is reversible and, in fact, is an unfavorable equilibrium, because the product (a simple ketone enolate) is a less stable anion than is the doubly stabilized malonate anion. However, the next step, reaction with more malonic ester to make a new malonate anion, drives the equilibrium to product. The reaction is catalytic in base because malonate is regenerated in this last step.

38. These reactions form molecules that have the appearance of products of nucleophilic substitution.



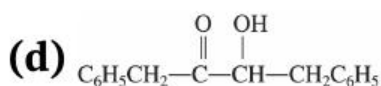
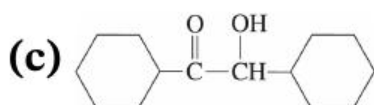
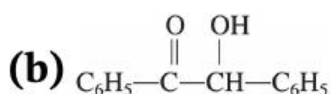
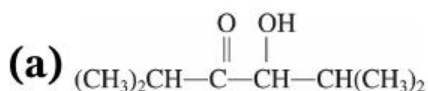


40. Work backward. Note carbon-carbon bonds being formed (arrows).

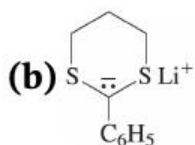
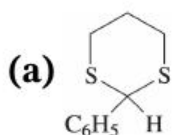


- (c) A sequence identical to that of (b), but substitute two alkylations with $\text{BrCH}_2\text{COCH}_3$ for the two Michael additions to $\text{H}_2\text{C}=\text{CHCOCH}_3$.
- $\text{H}_2\text{C}=\text{CHCOCH}_3$.

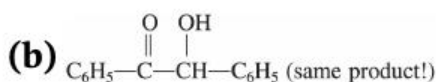
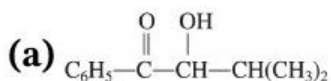
11.

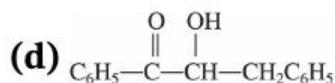
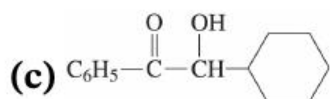


12.



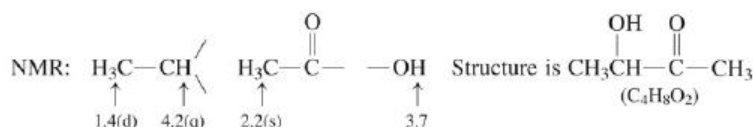
In the same order as in [Problem 41](#):





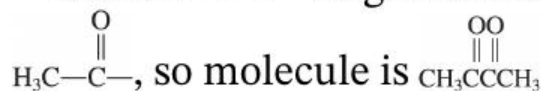
13.

- (a) A IR spectrum: Carbonyl and hydroxy groups (cannot be an amine because molecular weight is an **even** number).



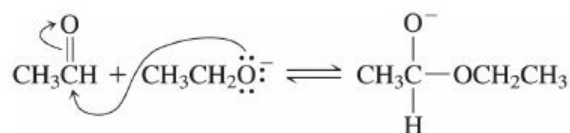
B Molecular weight is reduced by 2 units, so formula is probably now $C_4H_6O_2$. IR: Carbonyl signal only. NMR: All H's equivalent. MS: Molecule breaks in half readily, giving m/z 43, m/z 43,

C_2H_3O fragments. Simplest interpretation:

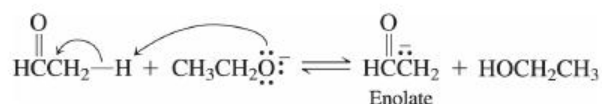


- (b) Oxidation. Churning cream mixes it with air, thus allowing O_2 to react with ketoalcohol A to make diketone B.
- (c) You can synthesize A by reaction of acetaldehyde with catalytic *N*-dodecyl-thiazolium salt (Section 23-4). Oxidation gives B.
- (d) The diketone is conjugated.

14. Addition to carbonyl:

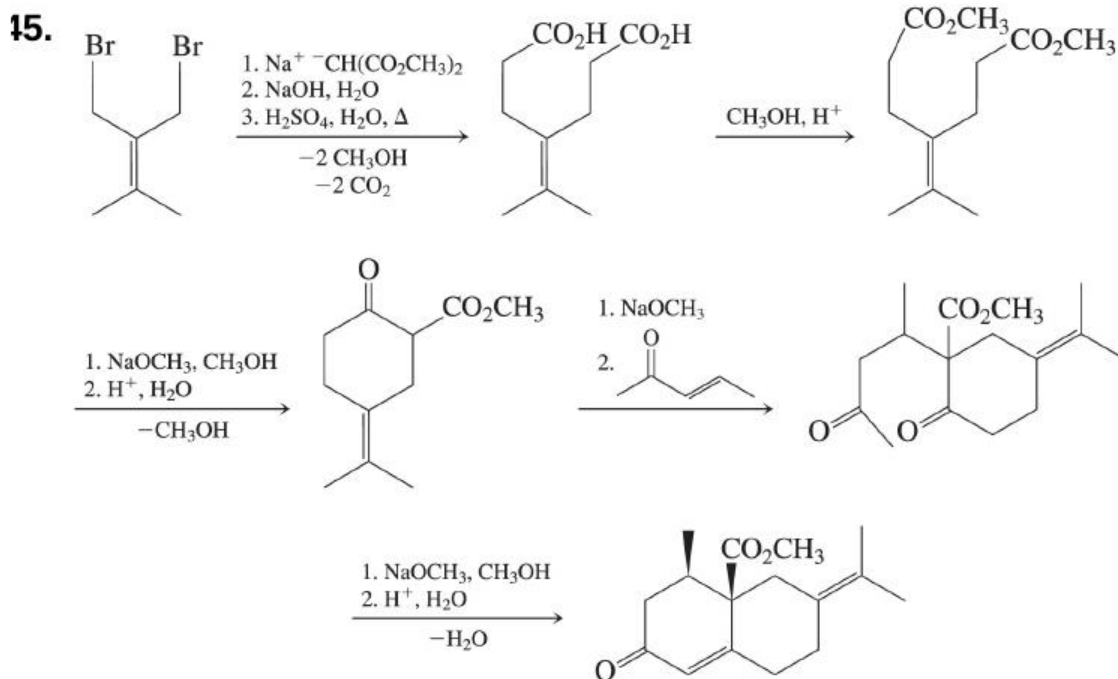


Deprotonation of α -carbon: α -carbon :



Deprotonation of the aldehyde carbon leads to a much poorer anion than the enolate: $\text{CH}_3\text{C}=\text{O}^-$, which has an electron pair in an sp^2sp^2 orbital, unable to be stabilized by resonance. Given the two favorable processes shown above,

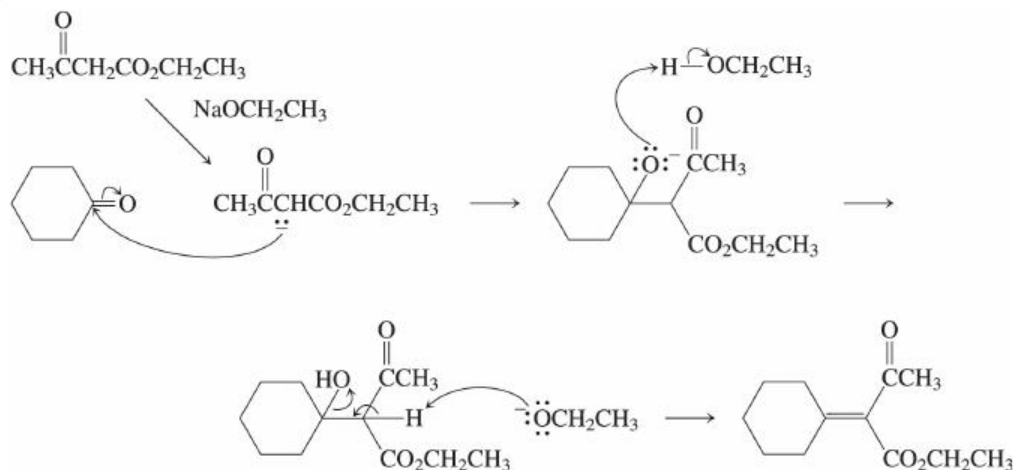
deprotonation of the —CH=O group is simply **not competitive**.



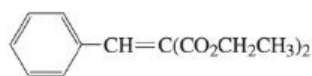
16. The Knoevenagel condensation is a variation of aldol condensation

using a β - β -dicarbonyl compound as the source of the enolate reaction partner. Its mechanism is the same as that of the aldol:

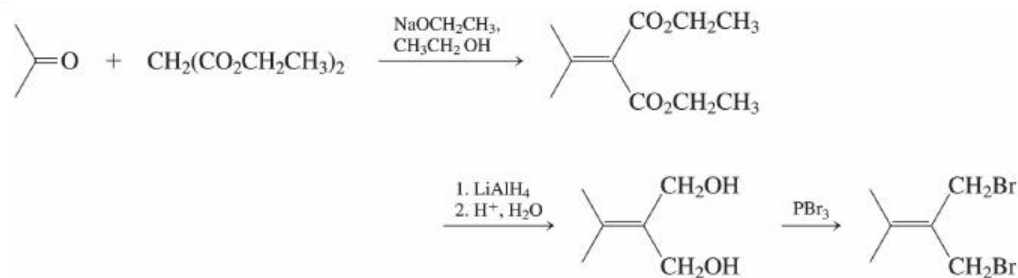
(a)



(b) As in the aldol, remove the elements of water (two hydrogen atoms from the carbon of the malonate ester and the oxygen atom from the aldehyde) and replace them with a carbon–carbon double bond:



(c)

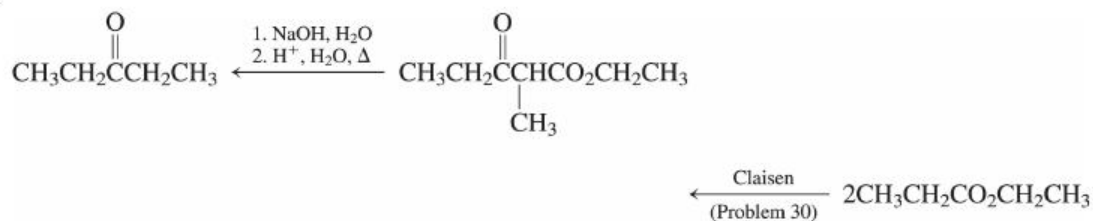


17. The acetoacetic ester ketone synthesis is good only for **methyl** ketones.

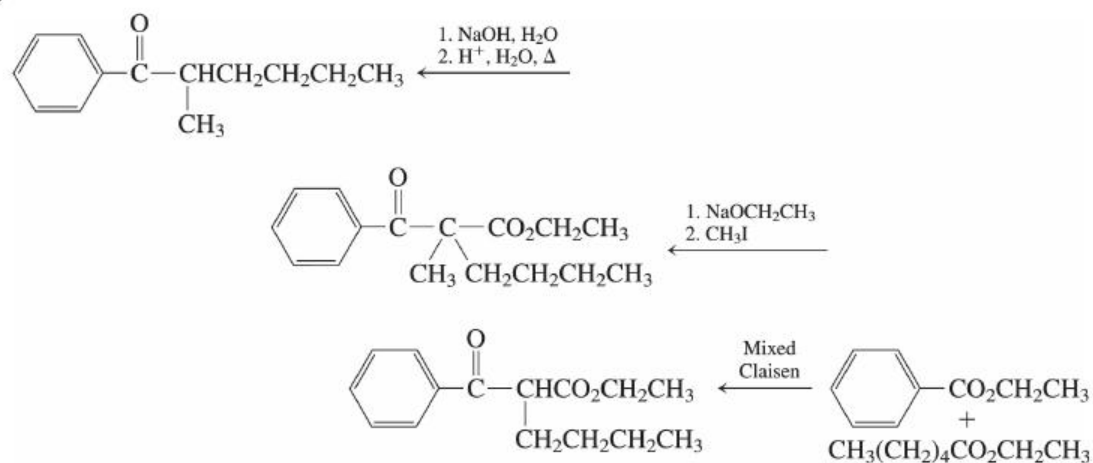


For other ketones, the appropriate 3-ketoester must be prepared by using a Claisen condensation.

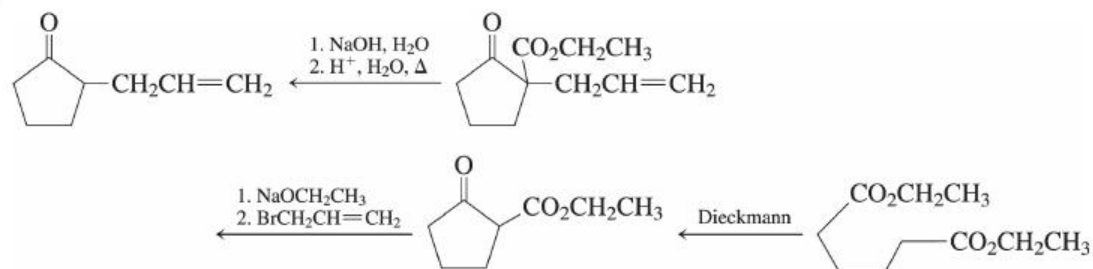
(a)



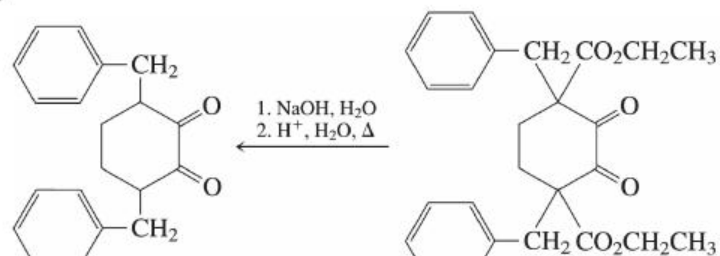
(b)

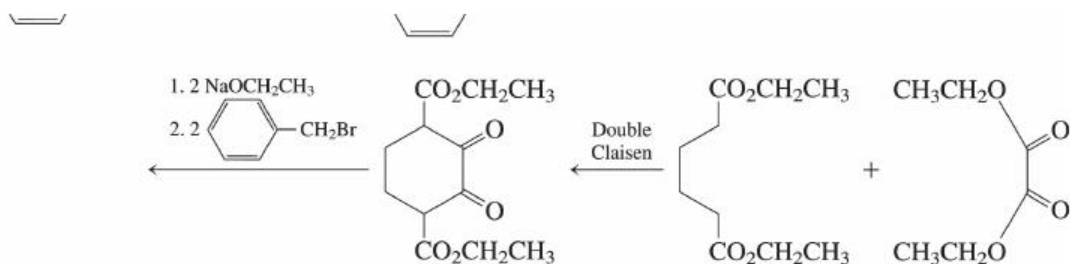


(c)



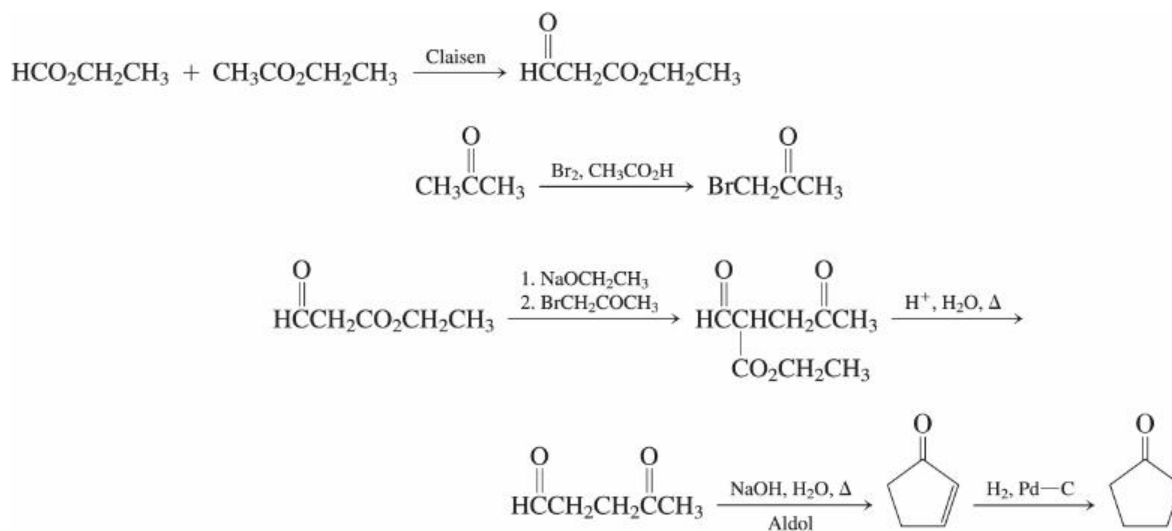
(d)



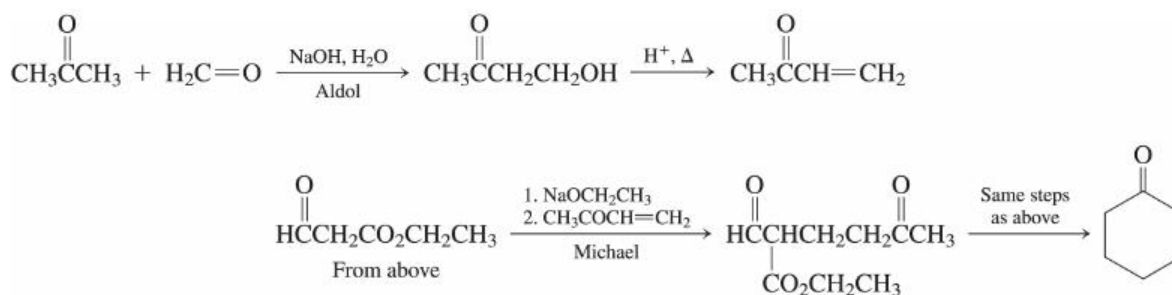


18. Ethanol is the reaction solvent unless specified otherwise.

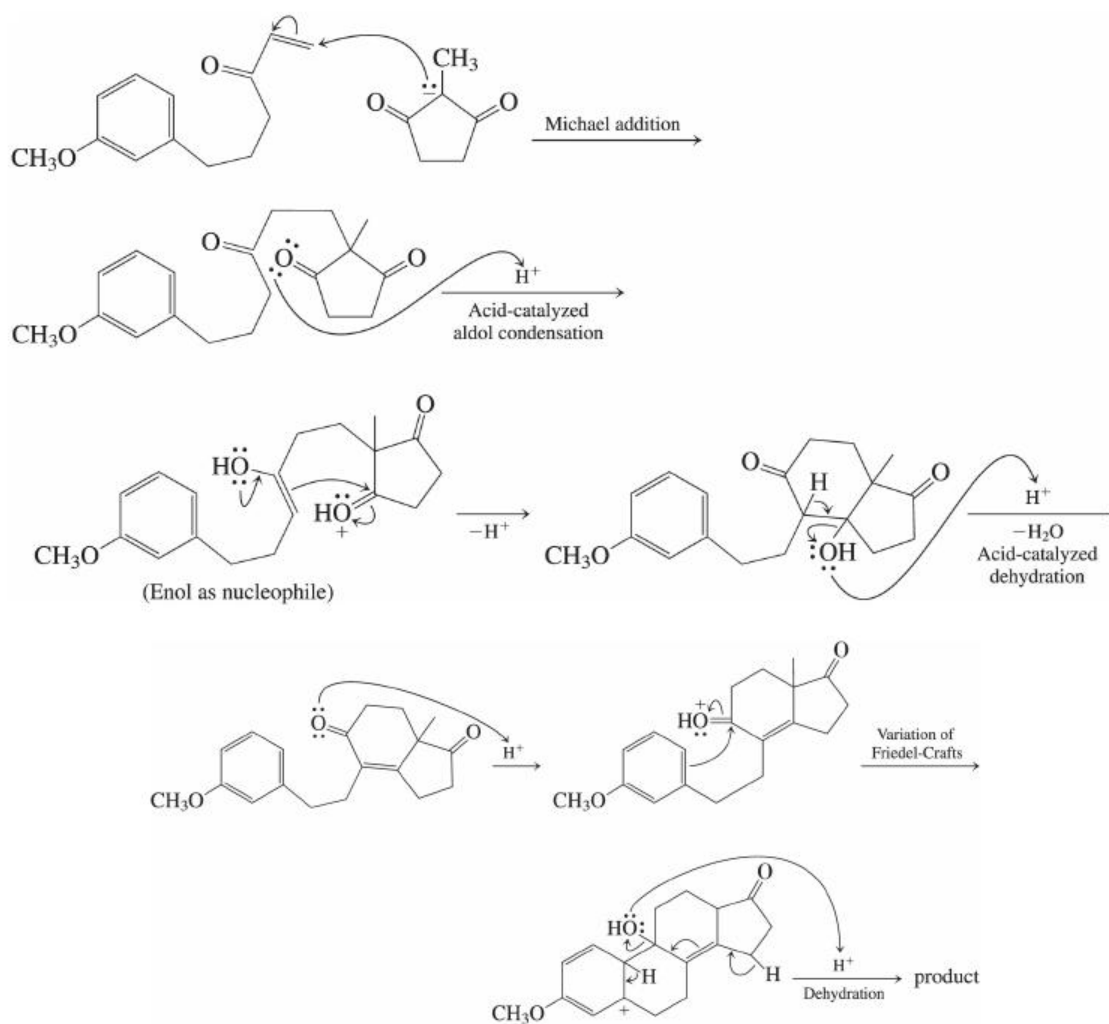
Cyclopentanone



Cyclohexanone

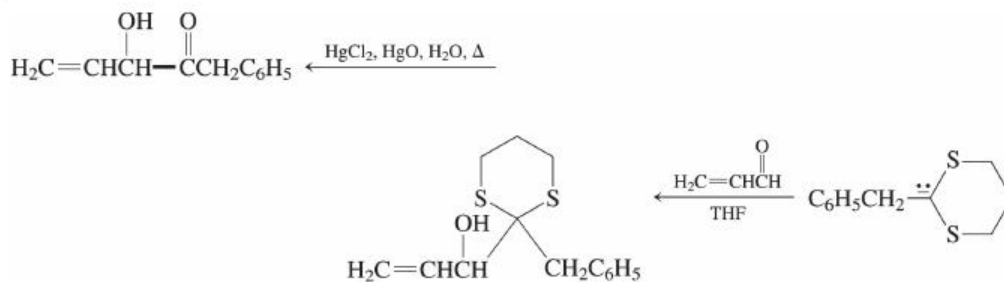


19.

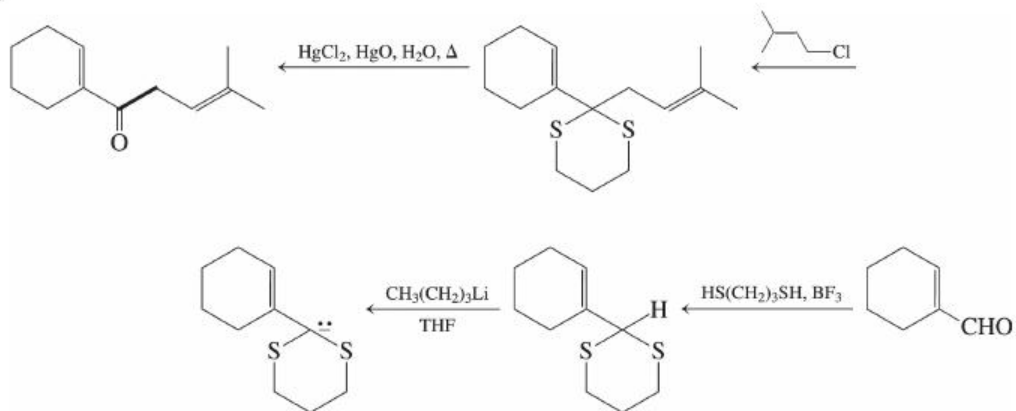


50. These are not easy. If you didn't get them, look at the answer for (a) and then try (b) and (c) again on your own. Retrosynthetic disconnections are indicated in boldface.

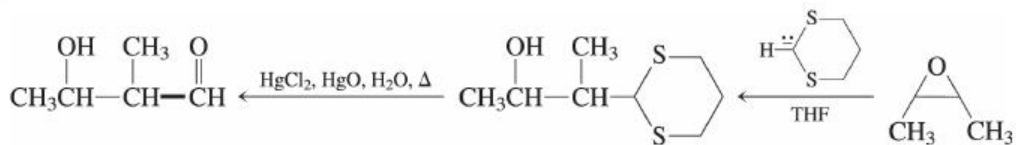
(a)



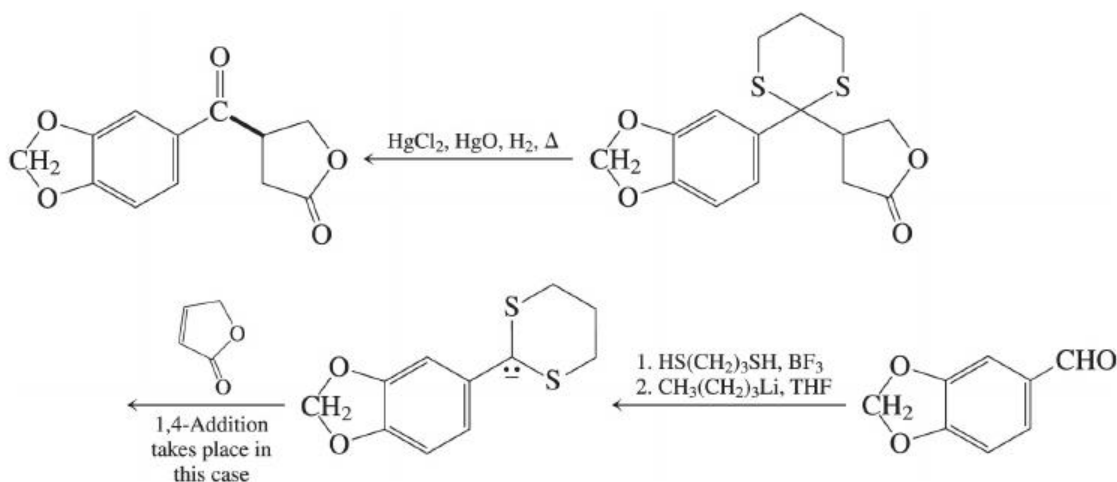
(b)



(c)

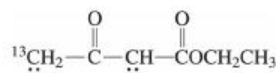


- 51.** Identify the bond being made. It looks like the result of 1,4-addition to the α,β -unsaturated lactone by an acyl anion equivalent.



52.

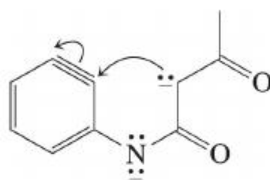
- (a) Notice that the sequence begins by treating the starting material with *two* equivalents of strong base. The result is the formation of a dianion, which can be represented by the following Lewis structure.



Of the two anionic carbon atoms, the terminal one (^{13}C) (^{13}C) is the more basic and therefore the stronger nucleophile because the charge is stabilized by only one adjacent carbonyl group. The remainder of the sequence is the completion of a β - β -ketoester ketone synthesis.

- (b) Attempted alkylation of a β - β -dicarbonyl anion (an $\text{S}_{\text{N}}2$ process) with a tertiary haloalkane is doomed to give E2 elimination instead (textbook, Section 7-9).

Extra problem: The key is the need for *three* equivalents of strong base. The first two deprotonate the β - β -ketoamide CH_2CH_2 at C2 and the NH, respectively. What does the third do? Elimination of HCl from the benzene ring to give a benzyne, to which the ketoamide carbanion adds:



53. (c)
54. (b)
55. (e) From heat-induced dehydration of the open-chain diacid to give the cyclic anhydride (see text [Section 19-8](#)).
56. (e) The IR band at 2250 cm^{-1} is a dead giveaway for the nitrile triple bond. This molecule is the only one that matches the NMR spectrum (a 2H singlet and an ethyl group).

24

Carbohydrates: Polyfunctional Compounds in Nature

In this chapter you'll begin to apply the material you've just seen to a major class of "real world" molecules, carbohydrates (sugars). For a change, nomenclature will play a more central role: The names of sugars and sugar derivatives follow their own independent system, which includes some special stereochemical terms. On the other hand, most of the reactions are old ones and are needed for only a limited number of purposes such as structure determination, interconversion of derivatives, and synthesis of one sugar from another. You need to be good at deductive reasoning so that you can solve the puzzles posed by some of the problems. If you can, then this chapter should not be too hard for you.

Outline of the Chapter

[24-1, 24-2, 24-3 Names and Structures of Carbohydrates](#)

Be prepared for a lot of new terminology.

[24-4 through 24-8 Polyfunctional Chemistry of Sugars](#)

The basics: mostly (but not entirely) review material.

[24-9, 24-10 Step-by-Step Buildup and Degradation of Sugars](#)

Application in synthesis and structure determination.

[24-11, 24-12 Complex Sugars in Nature](#)

Keys to the Chapter

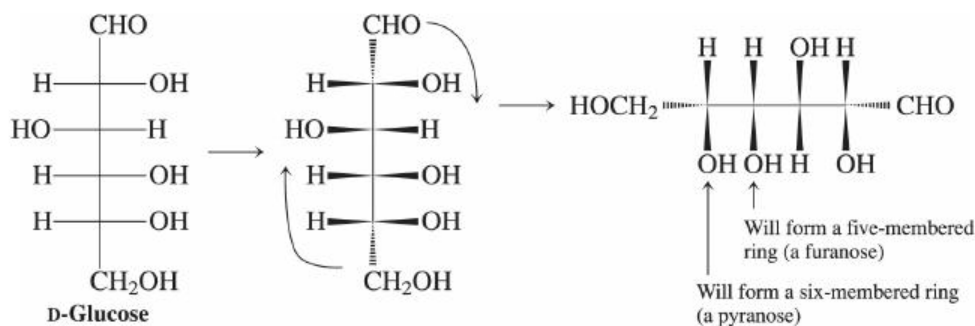
24-1. Names and Structures of Carbohydrates

The naming system presented in this section comes from an assortment of historically derived common names organized into a semiofficial framework that is universally used. So, sugars are all ketones or aldehydes containing alcohol groups, and their names all end in *-ose*. Almost all of them have stereocenters; and they are usually drawn in Fischer projections, with the carbon chain vertical and the carbonyl group nearest the top. If the stereocenter **nearest the bottom** of the Fischer projection has the **OHOH** on the right, it has an *RR* configuration, and the sugar is said to belong to the *D* family. If this **OHOH** is on the left, you have an *SS* configuration and an *L*-sugar. Look at Figures 24-1 and 24-2. Each horizontal row contains structures that are all **diastereomers** of one another. None of the mirror-image structures (**enantiomers**) of any of these is illustrated; the mirror image of any *D*-sugar is just an *L*-sugar with the same name and all the stereocenters reversed.

24-2 and 24-3. Conformations and Cyclic Forms of Sugars; Mutarotation

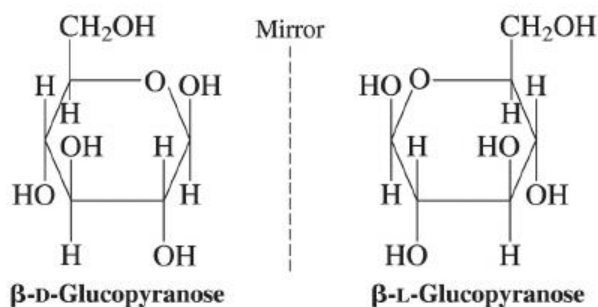
Because sugars contain alcohols and carbonyl groups, they can (and usually do) form cyclic hemiacetals, typically with either five- or six-membered rings. Translating an open-chain structure into a picture of the cyclic hemiacetal is tricky: Here's a step-by-step way to do it.

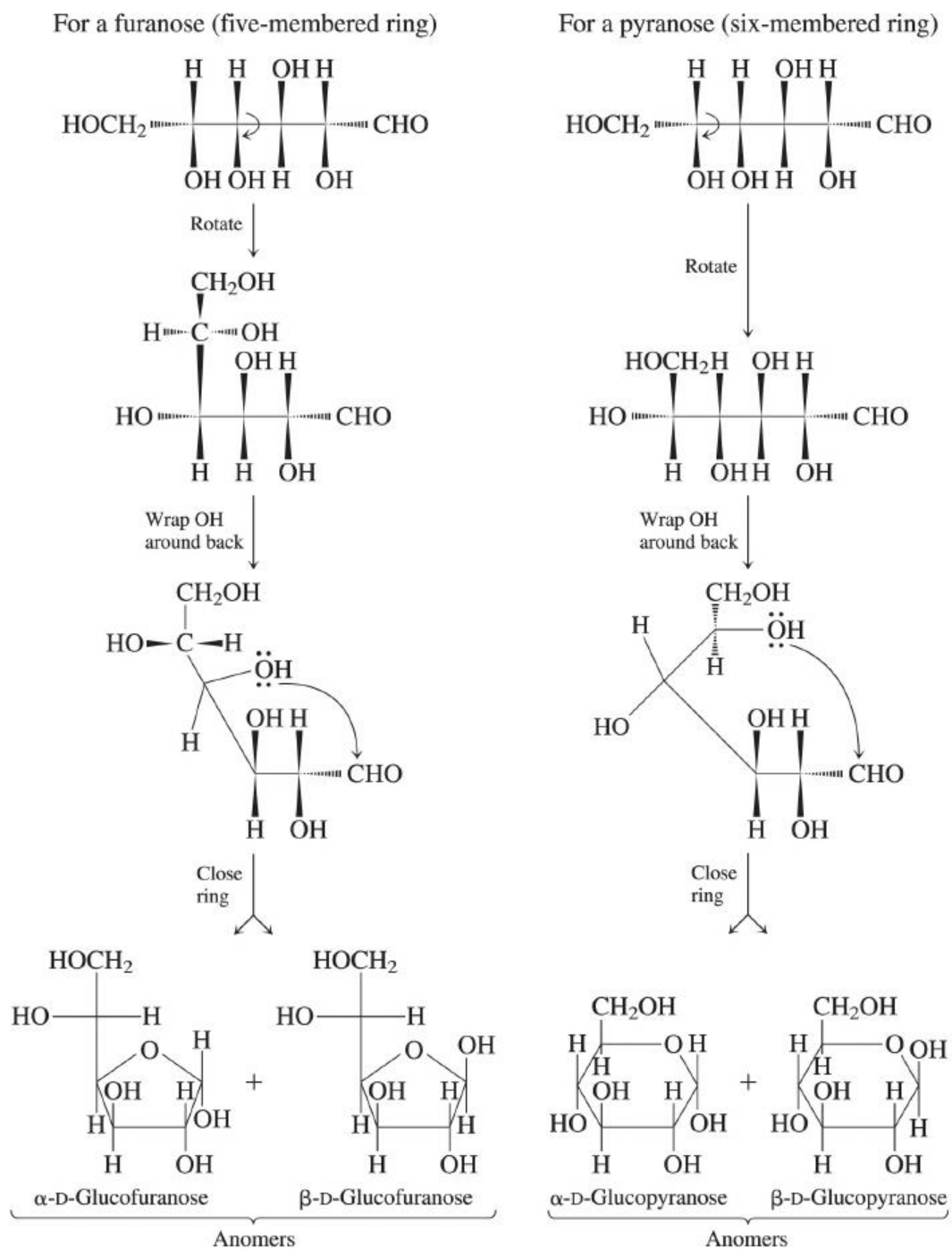
- .. Write in wedges and hashed lines and (for a *D*-series sugar) lay the structure on its side, with the top moved down to the right (a 90°90° clockwise rotation). Then, locate the **OHOH** group that you will use to form the cyclic hemiacetal with the carbonyl group.



1. Rotate around the C—C bond to the right of the OH group you picked out until the OH is horizontal and pointing away from you. (Make a model!) Then wrap this left-hand end of the chain behind the plane of the paper to put the OH near to the carbonyl carbon. Finally, make the hemiacetal bond. (This sequence is shown step-by-step on the next page.)

By convention, the procedure for L-series sugars is modified such that the top of the structure is rotated down **to the left** when it is laid on its side. That allows the L and D structures to look like mirror images when placed side-by-side, and preserves the proper stereochemical designations for the anomeric hydroxy groups: α being down and β being up.





If you like, you can derive the cyclic structure of an L-sugar by doing the D-sugar first and then writing its mirror image.

Remember, sugars in solution typically exist as equilibrium mixtures of open chain plus cyclic hemiacetal structures. For glucose, this

equilibrium mixture contains 63.6% β -pyranose, 36.4% α -pyranose, together with just traces of open chain and furanose structures. The interconversion of β and α anomers is called *mutarotation*.

24-4 through 24-8. Polyfunctional Chemistry of Sugars

Although most of the reactions in these sections are old ones, typical of either alcohol or aldehyde/ketone chemistry, a couple of new reagents are introduced. These mainly allow selective reactions to be carried out on the multifunctional sugar molecule. Examples are Br_2 in H_2O , which oxidizes only the aldehyde group of an aldose to CO_2H , and HNO_3 , which oxidizes both the end carbons of an aldose, forming a dicarboxylic acid. The reactions in these sections have been chosen for their importance in practical aspects of sugar chemistry, the material in the latter half of this chapter.

24-9 and 24-10. Step-by-Step Buildup and Degradation of Sugars

Determining sugar structures was a major effort involving the development of reaction sequences to lengthen or shorten sugar chains and the use of some very clever logic dealing with the consequences of the stereochemistry of sugars and sugar derivatives. The “Fischer proof” illustrates the main techniques used. Note that the process repeatedly makes use of the synthesis of dicarboxylic acids, which are tested for the presence or absence of optical activity. A sugar-derived optically inactive diacid is assumed to be a meso compound, containing a plane of symmetry, and this kind of information is used to narrow down possible structures for unknown sugars.

24-11 and 24-12. Complex Sugars in Nature

These sections present simple extensions of the material just completed.

Mother Nature has developed a convenient method for linking sugar molecules together: An alcohol group of one sugar forms an acetal by reactions with the hemiacetal group of another. This connection, called a **glycoside linkage**, is just a fancier version of a simple acetal formed by reaction of a sugar with an ordinary alcohol, like methanol (to form a “methyl glycoside,” as in [Section 24-8](#)). Again, the determination of unknown structures represents a common type of problem. There are two important features that distinguish sugars containing free hemiacetal groups from those lacking them. In solution, hemiacetals are always in equilibrium with aldehydes. So, hemiacetal-containing sugars (1) undergo mutarotation and (2) are readily oxidized by mild oxidants like Ag^+ . Ag^+ . The latter feature is the basis for the Tollens’s test for “reducing” (that is, oxidizable) sugars. Some of the sugars in this section contain acetal but not hemiacetal groups. Find them! These are examples of “nonreducing” sugars.